tf.bitcast Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/bitcast

> *Bitcasts a tensor from one type to another without copying data.*

Function Signatures

```
tf.bitcast( input: Annotated[Any, TV_Bitcast_T], type:
TV_Bitcast_type, name=None ) -> Annotated[Any,
TV_Bitcast_type]
```

Code Examples

```
tf.bitcast(
input: Annotated[Any, TV_Bitcast_T], type: TV_Bitcast_type,
name=None
) -> Annotated[Any, TV_Bitcast_type]
```


```
tf.bitcast(
input: Annotated[Any, TV_Bitcast_T], type: TV_Bitcast_type,
name=None
) -> Annotated[Any, TV_Bitcast_type]
```


```
a = [1., 2., 3.]
equality_bitcast = tf.bitcast(a, tf.complex128)
Traceback (most recent call last):
InvalidArgumentError: Cannot bitcast from 1 to 18 [Op:Bitcast]
equality_cast = tf.cast(a, tf.complex128)
print(equality_cast)
tf.Tensor([1.+0.j 2.+0.j 3.+0.j], shape=(3,), dtype=complex128)
```

```
a = [1., 2., 3.]
```

```
equality_bitcast = tf.bitcast(a, tf.complex128)
```

```
Traceback (most recent call last):
```

```
InvalidArgumentError: Cannot bitcast from 1 to 18 [Op:Bitcast]
```

Documentation

Bitcasts a tensor from one type to another without copying data.

View aliases

Compat aliases for migration

See
Migration guide for
more details.

tf.compat.v1.bitcast

Given a tensor input, this operation returns a tensor that has the same buffer data as input with datatype type.

If the input datatype T is larger than the output datatype type then the shape changes from [...] to [..., sizeof(T)/sizeof(type)].

If T is smaller than type, the operator requires that the rightmost dimension be equal to sizeof(type)/sizeof(T). The shape then goes from [..., sizeof(type)/sizeof(T)] to [...].

tf.bitcast() and tf.cast() work differently when real dtype is casted as a complex dtype
(e.g. tf.complex64 or tf.complex128) as tf.cast() make imaginary part 0 while tf.bitcast()
gives module error.
For example,

Example 1:

Example 2:

Example 3:

Returns